2.6 Compound interest

In practice, interest is often compounded several times a year. Suppose the interest is compounded m times a year, and the nominal interest rate is r. Then the interest rate per period is r/m. After t years, the accumulated amount $A(t)$ and the initial principal amount P are related by

$$A(t) = P\left(1 + \frac{r}{m}\right)^{mt}.$$

Example Suppose that the principal amount $P = \$1000$ is invested at the nominal rate of 6%. Find the accumulated amount after 5 years, if the interest is compounded
(a) quarterly, (b) monthly.

2.6 Compound interest

We have $r = 0.06$ and $t = 5$.

For part (a), we have $m = 4$ and hence

$$A(5) = 1000\left(1 + \frac{0.06}{4}\right)^{4\times 5} = 1000 \times 1.015^{20}$$
$$= 1000e^{20\ln 1.015} \doteq 1346.86.$$

For part (b), we have $m = 12$ and hence

$$A(5) = 1000\left(1 + \frac{0.06}{12}\right)^{12\times 5} = 1000 \times 1.005^{60}$$
$$= 1000e^{60\ln 1.005} \doteq 1348.85.$$

Effective Rate

The *effective rate* r_{eff} is the simple interest rate which would produce the same accumulated amount in 1 year as the nominal rate compounded m times per year. Thus

$$P(1+r_{eff}) = P\left(1+\frac{r}{m}\right)^m \Longrightarrow r_{eff} = \left(1+\frac{r}{m}\right)^m - 1.$$

Example Find the effective rate of interest corresponding to a nominal rate of 12% if the interest is compounded
(a) semiannually, (b) quarterly, (c) monthly.
Solution (a) In this case $m = 2$, so
$r_{reff} = \left(1+\frac{0.12}{2}\right)^2 - 1 = 0.1236.$

Effective Rate

(b) In this case $m = 4$, so $r_{reff} = \left(1 + \frac{0.12}{4}\right)^4 - 1 = 0.1255$.
(c) In this case $m = 12$, so $r_{reff} = \left(1 + \frac{0.12}{12}\right)^{12} - 1 = 0.1268$.
We note the following formula:

$$A = P(1 + r_{eff})^t = P\left(1 + \frac{r}{m}\right)^{mt}.$$

Example An investment firm offers two types of returns: the first one at the nominal rate of 25% compounded annually, and the second one at the nominal rate of 24% compounded monthly. Compare the effective rates of the two types and state which one pays more.

Effective Rate

Solution For the first one,

$$r_{eff} = 0.25.$$

For the second one,

$$r_{eff} = \left(1 + \frac{0.24}{12}\right)^{12} - 1 \approx 0.268.$$

The second one pays more money.

Present and Future Values

In the formula

$$A = P\left(1 + \frac{r}{m}\right)^{mt}, \text{ or } P = A\left(1 + \frac{r}{m}\right)^{-mt},$$

P is called the *present value* and A is the *future value*.

Example Suppose \$10000 is to be received in 5 years, and the nominal rate is 5.5% compounded annually. What is the present value?

Solution We have

$$P = 10000\,(1 + 0.055)^{-5} \approx 7,651.$$

The present value is \$7,651.

Effective Rate

Example Suppose the nominal rate is 0.1. What is the effective rate if the interest is compounded
(a) daily, (b) hourly ? (Assume 365 days/year)
Solution We have

$$(a)\quad r_{eff} = \left(1 + \frac{0.1}{365}\right)^{365} - 1 \approx 0.1052,$$

$$(b)\quad r_{eff} = \left(1 + \frac{0.1}{365 \times 24}\right)^{365\times 24} - 1 \approx 0.1052.$$

What if the interest is compounded *continuously* ? That is, what happens to $\left(1 + \frac{0.1}{m}\right)^m$ when m approaches ∞?

Effective Rate

This is related to the concept of the *limit* of a function. We will show that

$$\lim_{m\to\infty}\left(1+\frac{0.1}{m}\right)^m = e^{0.1} \doteq 1.1052.$$

If we set $x = 0.1/m$, then $m \to \infty$ is equivalent to $x \to 0+$ (x approaches 0 from the right side), and

$$\left(1+\frac{0.1}{m}\right)^m = (1+x)^{\frac{0.1}{x}} = e^{0.1\frac{\ln(1+x)}{x}},$$

$$\lim_{m\to\infty}\left(1+\frac{0.1}{m}\right)^m = \lim_{x\to 0+} e^{0.1\frac{\ln(1+x)}{x}}.$$

3.1 The limit of a function at a point

Now we study the values of the function $\dfrac{\ln(1+x)}{x}$ as x approaches 0 from the right side.

x	$\frac{\ln(1+x)}{x}$
0.01	0.995033
0.001	0.999500
0.0001	0.999950
0.00001	0.999995

It is clear that as x approaches 0 from the right side, $\frac{\ln(1+x)}{x}$ approaches 1. We write

$$\lim_{x\to 0+} \frac{\ln(1+x)}{x} = 1.$$

3.1 The limit of a function at a point

The same can be said when x approaches 0 from the left side, as shown in the next table.

x	$\frac{\ln(1+x)}{x}$
-0.01	1.00503
-0.001	1.00050
-0.0001	1.00005
-0.00001	1.00001

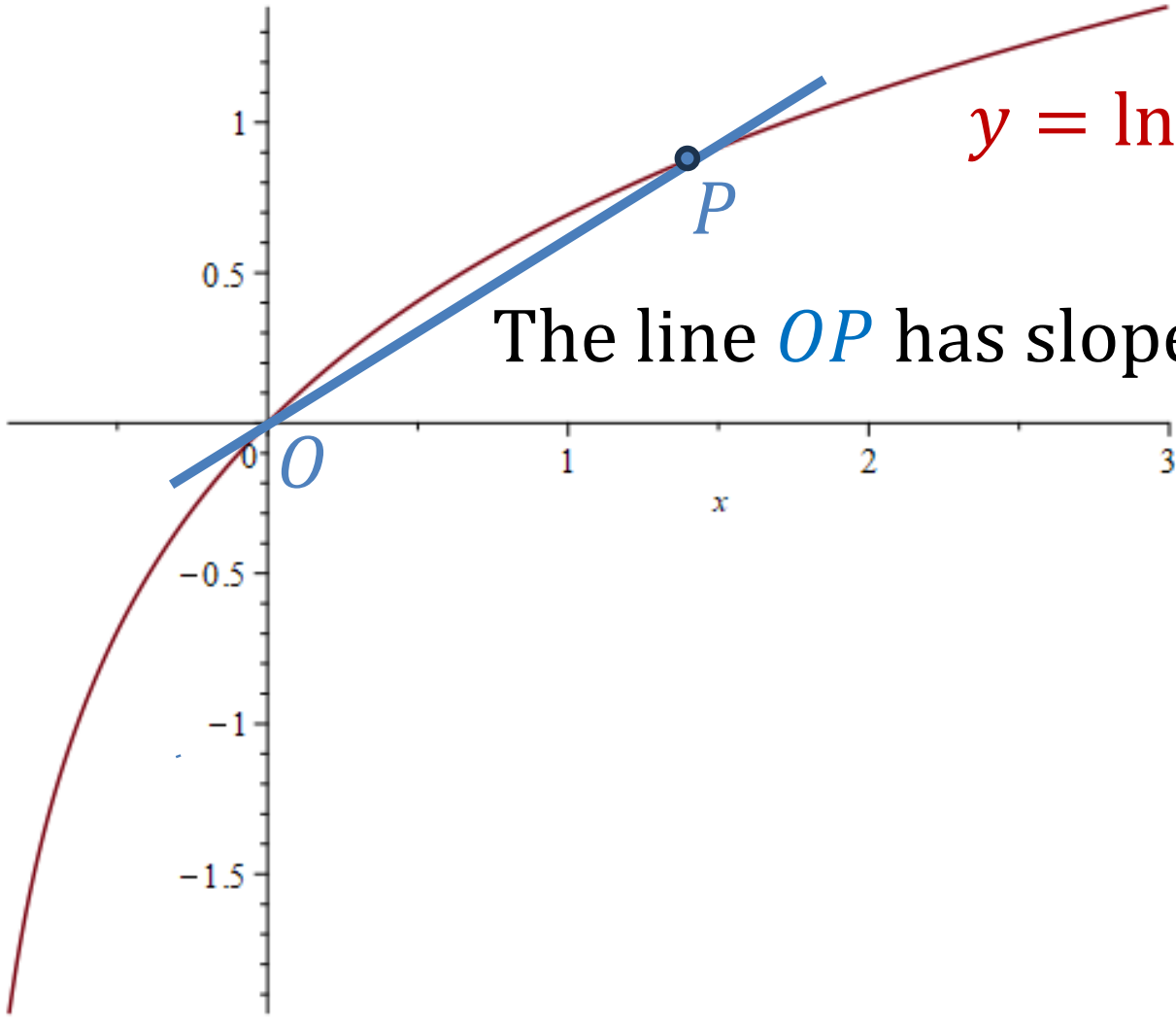
We write

$$\lim_{x \to 0-} \frac{\ln(1+x)}{x} = 1.$$

Since $\frac{\ln(1+x)}{x}$ approaches 1 as x approaches 0 from **both sides**, we write

$$\lim_{x \to 0} \frac{\ln(1+x)}{x} = 1.$$

3.1 The limit of a function at a point

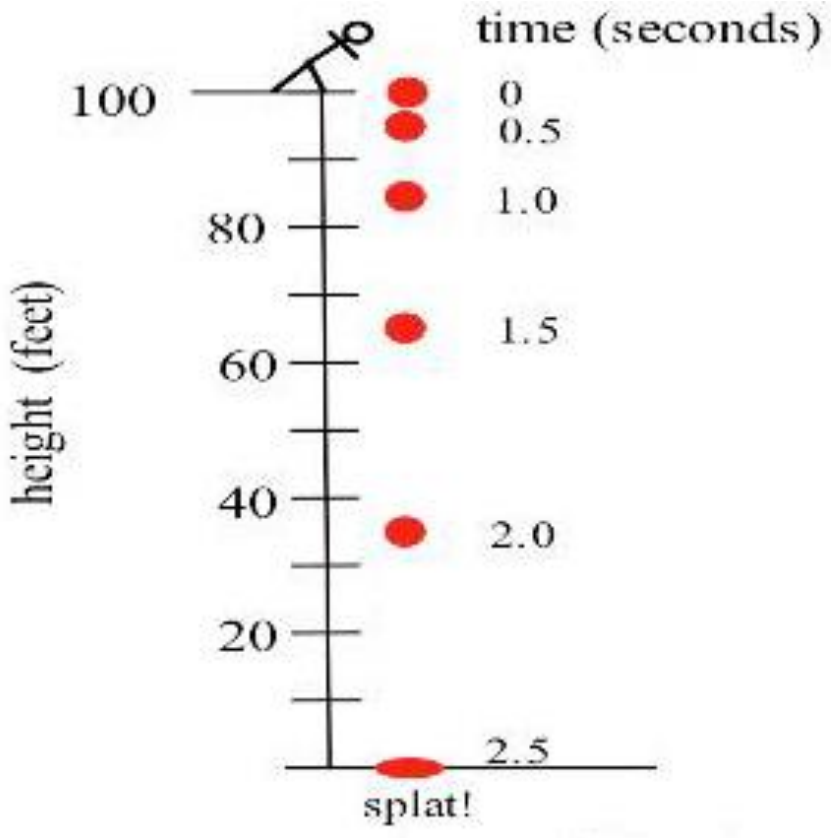


$y = \ln(1 + x)$

The line OP has slope $\dfrac{\ln(1+x)}{x}$

3.1 The limit of a function at a point

Recall the example about the falling tomato. How to find the instantaneous velocity of the tomato at a given time, say $t = 1$?



time (s)	height (ft)
0	100
0.5	96
1.0	84
1.5	64
2.0	36
2.5	0

3.1 The limit of a function at a point

We may use the *average velocity* in a small time interval around $t = 1$ to estimate the *instantaneous velocity* at $t = 1$. For example,

$$\frac{h(1.5) - h(1)}{1.5 - 1} = \frac{64 - 84}{0.5} = -40,$$
$$\frac{h(0.5) - h(1)}{0.5 - 1} = \frac{96 - 84}{-0.5} = -24.$$

The instantaneous velocity at $t = 1$ is approximated by the quotient $\frac{h(t)-h(1)}{t-1}$. As t approaches 1, the quotient approaches the instantaneous velocity of -32 (ft/s), as shown in the next table.

3.1 The limit of a function at a point

t near 1	$\frac{h(t)-h(1)}{t-1}$
1.1	-33.6
1.01	-32.16
1.001	-32.016
0.9	-30.4
0.99	-31.84
0.999	-31.984

Using

$$h(t) = 100 - 16t^2.$$

we will show that

$$\lim_{t \to 1} \frac{h(t) - h(1)}{t - 1} = -32.$$